

MODEL QUESTION PAPER

FIFTH SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING

MASS TRANSFER OPERATIONS (4091)

Maximum Marks:-100

Time:3hrs

PART-A

I Answer the following questions in one or two sentences (Each question carries two marks)

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|---|---------------------------------|---|
| 1 | Define diffusion | 2 |
| 2 | Define crystallisation | 2 |
| 3 | What is meant by bond water? | 2 |
| 4 | Define More volatile component. | 2 |
| 5 | What is meant by weeping? | 2 |

PART- B

- Answer any five question
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|----|---|---|---|
| II | 1 | Illustrate the constructional details and working of packed column | 6 |
| | 2 | Explain the constructional details and working of Swenson walker crystalliser | 6 |
| | 3 | Enumerate the industrial application of crystallisation. | 6 |
| | 4 | Derive an equation for 'q' line | 6 |
| | 5 | Illustrate the equilibrium distillation. | 6 |
| | 6 | Enumerate the application of leaching | 6 |
| | 7 | Ex plain the constructional details and working of Kennedy extractor | 6 |

PART -C

ANSWER ONE FULL QUESTION FROM EACH MODULE

Module I

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|-----|-----|---|---|
| III | (a) | Compare gas absorption and distillation | 8 |
| | (b) | Explain the mechanism of gas absorption | 7 |

OR

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|----|-----|---|---|
| IV | (a) | Explain the principle of ion exchange | 7 |
| | (b) | Differentiate between physical adsorption and chemical adsorption | 8 |

Module II

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|---|-----|--|---|
| V | (a) | Explain the constructional details and working of Krystal crystalliser | 8 |
| | (b) | Explain the constructional details and working of tray dryer | 7 |

OR

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|----|-----|---|---|
| VI | (a) | Derive an equation for calculating time for drying | 7 |
| | (b) | Illustrate constructional details and working of cooling crystalliser | 8 |

Module III

- | | | | |
|-----|-----|--|---|
| VII | (a) | Explain the mechanism of fractionation | 7 |
| | (b) | Vapour pressure of N-Heptane (A) and N-Octane (B) are given in the table. Assume that Raoult's law. Compute vapour –liquid equilibrium data at constant pressure of 101.325KPa and plot the boiling point diagram and Equilibrium diagram. | 8 |

TEMPERATURE	V.P of Heptane	VP of Octane
98.4	101.325	44.396
105	125.325	55.595
110	139.988	64.528
115	159.987	74.755
120	170.985	86.659
125.6	205.316	101.325

OR		
VIII	(a) A liquid mixture containing 40mole% methane and 60mole% water is fed to the differential distillation still at atmospheric pressure; with 60mole% of liquid is distilled. Find the composition of the composited distillate and the residue. DATA X = 0.05, 0.1, 0.2, 0.3, 0.4, 0.5 Y = 0.27, 0.42, 0.57, 0.66, 0.73, 0.78	8
	(b) Derive an equation for upper operating line or Rectifying line MODULE IV	7
IX	(a) Explain the constructional details and working of Bollmann extractor	8
	(b) Explain the triangular diagram for liquid-liquid extraction. OR	7
X	(a) Explain the counter current decantation system.	8
	(b) Explain the construction details and working of mixer settler	7